

HILLS COFFEE

Website Architecture & SEO Development Specification

Final implementation baseline: canonical directory model, page templates, entity data, crawl/indexation rules, structured data, internal linking engine, migration rules and developer acceptance criteria.

Build principle: Hills Coffee is a B2B green coffee sourcing platform; SEO pages are stable, server-rendered views of commercial buyer intent, coffee origins, offer-list inventory, knowledge content and trust entities.

1. Executive Summary

This specification defines the final build-ready website architecture for Hills Coffee. It converts the SEO strategy workbook into implementation rules that product, UX, development, content, SEO and QA teams can follow without reinterpreting keyword research or technical SEO requirements.

- Hills Coffee must be implemented as a B2B green coffee sourcing and trading platform, not a retail coffee shop architecture.
- Commercial supplier pages, origin pages, offer-list inventory, knowledge guides and trust pages each own a distinct search intent.
- All indexable pages must expose useful HTML, metadata, canonical tags, structured data and crawlable links before client-side enhancement.
- Internal linking must be generated from entity relationships: Origin ↔ Coffee Lot ↔ Commercial Page ↔ Knowledge Guide ↔ RFQ.
- Filters, app states, cart/wishlist and arbitrary query combinations must not create uncontrolled indexable URLs.
- Metadata and structured data must never output placeholder values such as [object Object].

1.1 Final Implementation Baseline

Architecture Area	Final Specification	Development Requirement
Site model	B2B green coffee sourcing platform for roasteries, cafés, factories, private labels, hotels, distributors and coffee traders.	Build only approved SEO routes. New indexable routes require SEO/product sign-off.
Commercial pages	Supplier, wholesale, specialty, raw/roastery and pricing pages own high-intent B2B queries.	Use reusable commercial templates with page-specific copy, RFQ CTAs and origin/offer-list links.
Origin architecture	/coffee-origins/ is the hub; /coffee-origins/{origin}/ owns country-origin commercial investigation.	Origin pages must connect to active lots, commercial pages and relevant guides.
Offer-list inventory	/green-coffee-offer-list/ is the canonical active inventory hub; lot detail URLs use stable IDs.	Do not rely on legacy /products/{id} as the long-term canonical surface.
Knowledge layer	/knowledge/ supports informational and commercial-investigation queries without competing with money pages.	Each guide has one primary commercial target and visible author/reviewer/dates.
Technical SEO	Server/raw HTML is the primary SEO delivery layer. JavaScript can enhance, not replace, core links/content.	Fix canonical mismatch, JS-only links, structured-data validation issues and image dimension/performance issues as launch blockers.

2. Architecture Principles

- One intent, one canonical URL: the same head keyword cluster must not be represented by multiple competing pages.
- Entity-first architecture: origins, coffee lots, commercial services, buyer segments, guides and organization/person entities are structured data objects, not isolated pages.
- B2B before ecommerce: users should understand supply, samples, quote workflow, quality and logistics before they see generic shop/cart behavior.
- Stable hubs, dynamic inventory: commercial and origin pages are durable; offer-list lots can appear/disappear and must follow lifecycle rules.
- HTML first: search engines and AI retrieval systems must see titles, H1s, main copy, tables, cards, links and schema without waiting for client-only rendering.

- Green coffee vs 1718 separation: Hills targets B2B green/raw/sourcing procurement; retail roasted coffee journeys should be clearly separated or routed away when needed.

Commercial Page → supports → RFQ

Origin → contains/supplies → Coffee Lots

Coffee Lot → belongsTo → Origin

Knowledge Guide → supports → one Commercial Page

Organization/Person → proves → Trust / Author / Contact

3. Canonical Sitemap & Directory Model

The following directory model is the authoritative architecture baseline. Routes are lowercase, hyphenated and use a trailing slash consistently.

```

/
├── green-coffee-beans-supplier/
├── coffee-beans-supplier/
├── wholesale-coffee-beans/
├── specialty-coffee-beans/
├── raw-coffee-beans-for-roasters/
├── bulk-coffee-beans/
├── coffee-beans-wholesale-price/
├── arabica-coffee-beans-wholesale/
├── robusta-coffee-beans-wholesale/
├── green-coffee-offer-list/
│   └── {origin-lot-slug}-{stable-id}/
├── coffee-origins/
│   ├── brazil/
│   ├── ethiopia/
│   ├── colombia/
│   ├── vietnam/
│   ├── uganda/
│   └── kenya/
├── knowledge/
│   ├── what-are-green-coffee-beans/
│   ├── green-coffee-vs-roasted-coffee/
│   ├── arabica-vs-robusta-commercial-buyers/
│   ├── how-to-choose-green-coffee-supplier/
│   ├── how-roasteries-source-coffee-beans/
│   ├── coffee-bean-grades-explained/
│   ├── coffee-defects-explained/
│   └── pallet-vs-container-coffee-buying/
├── request-a-quote/
├── about/
├── contact/
├── help/
├── shipping/
├── returns/
├── privacy/
├── cookies/
└── terms/
  
```

└─ cart/	[noindex]
└─ wishlist/	[noindex]

3.1 Utility routes

Route	Indexation	Purpose
/cart/	noindex, noarchive where appropriate	Checkout state; not an SEO landing page.
/wishlist/	noindex, noarchive where appropriate	Private/user saved lots; not an SEO landing page.
/?sort=, ?filter=, ?view=	noindex/follow or canonicalized to clean parent	User filter/sort states. Do not allow indexable parameter combinations unless approved.
/api/*, /admin/*, preview/staging	blocked or protected depending on environment	Internal systems, staging and API endpoints.

3.2 Page type distinctions

Layer	Example URL	Question Answered	Organic Intent
Commercial supplier page	/green-coffee-beans-supplier/	Can this company supply green coffee for my business?	green coffee beans supplier
Origin hub/detail	/coffee-origins/ethiopia/	Can Hills supply Ethiopian coffee? What profiles and grades exist?	ethiopian coffee beans
Offer list	/green-coffee-offer-list/	What lots are currently available?	green coffee offer list
Coffee lot detail	/green-coffee-offer-list/ethiopia-guji-natural-g4-hc-1001/	Can I request this exact lot?	specific lot / origin / grade
Knowledge guide	/knowledge/how-to-choose-green-coffee-supplier/	How should I evaluate a supplier?	commercial investigation

4. Organic B2B Visitor Journeys

1. Supplier discovery: Google / AI → /green-coffee-beans-supplier/ → origins / offer list → RFQ.
2. Wholesale price investigation: Google / AI → /coffee-beans-wholesale-price/ → /wholesale-coffee-beans/ → request latest price list.
3. Origin sourcing: Google / AI → /coffee-origins/ethiopia/ → available Ethiopia lots → sample request.
4. Roastery procurement: Google / AI → /raw-coffee-beans-for-roasters/ → sourcing process → current offer list → quote.
5. Knowledge support: Google / AI → /knowledge/coffee-bean-grades-explained/ → specs-based quote → RFQ.

5. Navigation Architecture

Primary Nav Item	Key Children / Destinations	SEO Role
Sourcing	Green Coffee Supplier; Coffee Beans Supplier; Wholesale; Specialty; Raw Coffee for Roasteries	Routes high-intent B2B procurement queries to durable commercial pages.
Origins	Coffee Origins hub; Brazil; Ethiopia; Colombia; Vietnam; Uganda; Kenya	Creates country-origin discovery and links to offer-list inventory.
Offer List	Current green coffee lots; filters; lot detail pages	Turns B2B sourcing demand into active inventory and sample/quote CTAs.
Knowledge	Green coffee basics; sourcing supplier guide; Arabica vs Robusta; grades; defects; pallet vs container	Captures informational/commercial-investigation demand and routes it to money pages.
About / Contact	About; Contact; Request a quote	Trust and conversion layer.

6. Page Template Contracts

ID	Template	Applies To	Purpose	Rendering	Indexation	Schema	Main Components
T01	Homepage	/	Brand entry + B2B coffee sourcing gateway	SSG/ISR	index	Organization + WebSite + WebPage	Hero, commercial pathways, origins, offer list, trust, knowledge, RFQ
T02	Commercial Supplier Landing Page	/green-coffee-beans-supplier/, /coffee-beans-supplier/, /wholesale-coffee-beans/, /specialty-coffee-beans/, /arabica-coffee-beans-wholesale/, /robusta-coffee-beans-wholesale/	B2B supplier/sourcing keyword ownership	SSG/ISR	index	WebPage + Service + BreadcrumbList + FAQ where visible	Hero, sourcing capability, origin proof, served segments, process, FAQ, RFQ
T03	Offer List Hub	/green-coffee-offer-list/	Active inventory / available green coffee lots	SSR or inventory-aware ISR	index if meaningful inventory exists	CollectionPage + ItemList + BreadcrumbList	Intro, filters, lot cards, origin pathways, RFQ guidance, pagination

ID	Template	Applies To	Purpose	Rendering	Indexation	Schema	Main Components
T04	Coffee Lot Detail	/green-coffee-offer-list/{origin-lot-slug}-{stable-id}/	Specific coffee lot / SKU investigation	SSR or inventory-aware ISR	index when active/unique; 404/410 when gone	Product + Offer when price/availability visible + BreadcrumbList	Gallery, lot facts, grade, process, quantity, documents, sample/quote CTA
T05	Origin Hub	/coffee-origins/	Origin discovery and commercial investigation	SSG/ISR	index	CollectionPage + ItemList + BreadcrumbList	Origin intro, origin cards, profiles, offer-list links, sourcing CTA
T06	Origin Detail	/coffee-origins/{origin}/	Country origin sourcing intent	ISR	index	CollectionPage/WebPage + Place + ItemList + BreadcrumbList + FAQ visible	Origin hero, regions, cup profile, grades, harvest, availability, related lots, FAQ, RFQ
T07	Buyer Segment Page	/raw-coffee-beans-for-roasters/ and future segment pages	Audience-specific procurement intent	SSG/ISR	index	WebPage + Service + BreadcrumbList	Segment hero, problems solved, sourcing model, quality workflow, offer list, CTA
T08	Commercial Guide / Pricing / Bulk Page	/coffee-beans-wholesale-price/, /bulk-coffee-beans/	Procurement/pricing/logistics commercial investigation	SSG/ISR	index	Article/WebPage + Service where applicable + BreadcrumbList	Quick answer, variables, tables, scenarios, CTA
T09	Knowledge Hub	/knowledge/	Informational cluster discovery	SSG/ISR	index	CollectionPage + ItemList + BreadcrumbList	Topic clusters, featured guides, author trust, commercial pathways
T10	Knowledge Detail	/knowledge/{slug}/	Informational / commercial support query	SSG/ISR	index	Article/BlogPosting + Person + Organization + BreadcrumbList	Quick answer, sections, examples, FAQ, author/reviewer, primary commercial target
T11	RFQ / Contact Page	/request-a-quote/, /contact/	Lead conversion and contact trust	SSG/ISR	index or noindex by business preference; default index for /contact and /request-a-quote	ContactPage + Organization + Service where applicable	Contact details, RFQ form, sourcing needs, trust, response expectations
T12	Trust / Entity Page	/about/	Brand, founder, company credibility	SSG/ISR	index	AboutPage + Organization	Company story, founder, trade

ID	Template	Applies To	Purpose	Rendering	Indexation	Schema	Main Components
							model, sustainability, trust signals, CTA
T13	Support / Legal Page	/help/, /shipping/, /returns/, /privacy/, /cookies/, /terms/	Support/compliance	SSG	index for support/legal if meaningful	WebPage; FAQPage optional only for visible support FAQ	Policy/help content, dates, contact details
T14	Utility App State	/cart/, /wishlist/, account/search/filter states	User app functionality	CSR allowed	noindex, follow for public utility states; noindex/noarchive for private states	None unless clearly applicable	Functional UI, canonical entity links where useful

6.1 Required section order by template

Template	Section Order
T01 Homepage	1 Header/navigation; 2 Hero + RFQ/search; 3 Commercial sourcing pathways; 4 Current offer-list preview; 5 Coffee origins; 6 Who Hills serves; 7 Quality/logistics proof; 8 Knowledge guides; 9 Trust/company evidence; 10 Final RFQ CTA; 11 Footer
T02 Commercial Supplier Landing	1 Header; 2 Breadcrumb; 3 H1 + answer + CTA; 4 Why Hills / business need; 5 Supply capabilities; 6 Origins available; 7 Quality control and grading; 8 Who we serve; 9 Sourcing process; 10 Related guides; 11 FAQ; 12 Final RFQ; 13 Footer
T03 Offer List Hub	1 Header; 2 Breadcrumb; 3 H1 + inventory explanation; 4 Filters/search; 5 Server-rendered lot cards; 6 Pagination; 7 Origin links; 8 How to request samples/quote; 9 FAQ; 10 CTA; 11 Footer
T04 Coffee Lot Detail	1 Breadcrumb; 2 Gallery/status; 3 H1; 4 Key facts; 5 Price/availability panel if factual; 6 Sample/quote CTA; 7 Description/cup profile; 8 Origin context; 9 Documents/specs; 10 Similar lots; 11 Contact; 12 Footer
T05 Origin Hub	1 Intro/H1; 2 Origin cards; 3 How to choose origin; 4 Featured active lots; 5 Related commercial pages; 6 Knowledge guides; 7 FAQ; 8 RFQ CTA
T06 Origin Detail	1 Breadcrumb; 2 Origin hero; 3 Regions and profile; 4 Common grades/processes; 5 Harvest/seasonality; 6 Active lots; 7 Commercial uses; 8 Quality/logistics; 9 Related origins/guides; 10 FAQ; 11 RFQ
T10 Knowledge Detail	1 Breadcrumb; 2 Title/H1; 3 Quick answer; 4 Author/reviewer/dates; 5 Table of contents; 6 Structured sections; 7 Tables/checklists; 8 Sources; 9 FAQ; 10 Primary commercial target; 11 Related guides

6.2 Shared Component Library

ID	Component	User Function	Development Function	Required Data	SEO Impact	Schema	Mobile Rule
C01	Global Header / Navigation	Move users to sourcing pages, origins, offer list, knowledge and RFQ	Server-render semantic nav with standard <a href> links	Priority URL list, active route	Distributes crawl equity to P0/P1 pages	WebSite/siteNavigationElement optional	Accessible menu, no hover-only navigation
C02	Breadcrumbs	Show hierarchy and parent pages	Generate from canonical route tree	Page parent, labels, canonical URLs	Reinforces hierarchy and feeds BreadcrumbList	BreadcrumbList	Visible, compact, touch-friendly
C03	Page Hero / Intent Block	Make page purpose obvious	Render H1, short answer, primary CTA in raw HTML	H1, summary, CTA, hero image	Confirms keyword-to-page relevance	WebPage.name/headline; Service.name	Keep above-fold copy short
C04	RFQ / Lead CTA	Convert procurement visitors	Accessible form/button; track events without hiding content	CTA text, form fields, destination	Converts organic visitors without blocking crawl	Action / ContactPoint where appropriate	Sticky CTA allowed if non-obstructive
C05	Origin Card	Route users by source country	Crawlable card to /coffee-origins/{origin}/	Origin name, country, profile, available lots	Builds origin cluster and crawl paths	ItemList ListItem / Place	Cards stack cleanly
C06	Coffee Lot Card	Preview a lot/offering	Crawlable anchor to lot detail; render facts in HTML	Lot name, origin, grade, process, availability	Connects offer hub/origins to inventory	ItemList / Product summary	Facts must remain readable
C07	Lot Key Facts Table	Make sourcing facts scannable	HTML definition list/table from inventory API/CMS	Origin, variety, process, grade, quantity, harvest, moisture, defects	Machine-readable procurement facts	Product/additionalProperty	Responsive stacked table
C08	Price / Availability Panel	Show quote/stock state	Use live data; never stale placeholders	Price if visible, currency, stock/availability	Feeds Offer only when visible/factual	Offer	Prominent but not blocking
C09	Origin Profile Content	Explain country/region sourcing value	Structured headings, tables, examples	Regions, cup profile, harvest, processes	Unique topical coverage by origin	Place / Article supporting facts	No long text wall above CTA
C10	Quality / QC Module	Prove sourcing reliability	Visible methods, sample process, documents	QC process, cupping, grading, sample rules	Supports trust and commercial conversion	Service/additionalProperty	Use concise steps
C11	FAQ / Q&A	Answer page-specific questions	Visible Q&A only; optional FAQPage semantic markup	Approved questions and answers	Long-tail support and AI extraction	FAQPage optional	Accordion content crawlable

ID	Component	User Function	Development Function	Required Data	SEO Impact	Schema	Mobile Rule
C12	Author / Reviewer Block	Show provenance for guides	Link to stable team/person profiles	Author, reviewer, dates, credentials	Strengthens article trust and entity clarity	Person + Article author/reviewer	Visible below or beside article
C13	Related Entity Links	Continue discovery naturally	Generate from entity relationships with limits	Origin, lots, pages, guides	Core internal linking engine	Referenced @id URLs	Bounded link count
C14	Pagination / Continuation	Expose deeper lots/guides	Crawlable paginated URLs; infinite scroll only enhancement	Page number, total, next/prev	Avoids orphaned inventory	CollectionPage/Item List	Buttons large enough
C15	Trust / Company Block	Expose business entity and proof	Single source of truth for name/contact/profiles	NAP, founder/company details, sourcing model	Organization consistency	Organization/Contact Point	Compact and clear

7. Entity & Data Model

Entity	Required Fields	Relationships	Indexation / Schema Critical Fields
Organization	id, legal_name, brand_name, logo, URL, phone, email, address, sameAs, founder/team, credentials, trading model	publishes guides, provides sourcing services, sells/offers coffee lots	name, URL, logo, contact details, sameAs, ContactPoint
Origin	id, canonical_name, slug, country, aliases, regions[], summary, cup_profile, processes[], grades[], harvest_calendar, media, market_data_date	has lots[], supports commercial pages, related guides	canonical name, slug, country/Place, profile facts, related active lots
Coffee Lot / Offer	stable_id, slug, title, origin_id, variety, process, grade, screen_size, quantity, availability, price if visible, currency, harvest/crop year, images, documents, timestamps	belongs to origin, classified by type/grade/process, links to RFQ	stable_id, status, title, origin, availability, price if visible, images
Commercial Page	id, slug, primary_keyword, H1, intent, target_audience, CTA, section content, FAQs, related origins/guides	links to origins, offer list, guides, RFQ	unique H1/title/meta/canonical, commercial target
Knowledge Guide	id, slug, title, intent, author_id, reviewer_id, dates, sources, FAQ,	supports one primary commercial page, links to related guides	author/reviewer/dates, canonical, Article schema

Entity	Required Fields	Relationships	Indexation / Schema Critical Fields
	primary_commercial_target, related_entities		
Person	id, name, role, credentials, languages, profile URL, contact if public	author/reviewer/consultant relationships	Person name, role, profile URL

8. URL, Canonical & Inventory Lifecycle Rules

- Preferred route style: lowercase, hyphenated, trailing slash enforced for public canonical routes.
- Canonical host: <https://www.hillscoffees.com/>. All http/non-preferred host variants redirect in one hop.
- Lot detail URL: </green-coffee-offer-list/{origin-lot-slug}-{stable-id}/>. Stable ID is the identity; descriptive slug may change.
- Changed lot slug for the same stable ID returns 301 to the active canonical slug.
- Aliases such as </origins/> or </products/> must not become competing indexable pages. Use one-hop 301 redirects to final canonical pages.
- Canonical tags must exist in raw/server HTML and match rendered HTML.

State	HTTP / SEO Behavior
Active commercial/origin/guide page	200; indexable; self-canonical; included in the correct XML sitemap.
Active coffee lot with unique useful content	200; indexable if product facts are sufficient; included in lot sitemap.
Lot sold out but useful archive retained	200 + noindex or index by policy; show status clearly; remove from active offer list if no longer available.
Changed lot slug with same stable ID	301 to current canonical lot URL.
Permanently removed lot with no replacement	404 or 410; remove from sitemap and internal links.
Invalid lot stable ID	404.
Temporary API/upstream failure	503 or controlled failure; never empty 200 “not found” page.

8.1 Faceted navigation and filters

URL / State	Behavior
/green-coffee-offer-list/	Indexable canonical inventory hub.
/green-coffee-offer-list/?origin=ethiopia&process=natural	noindex, follow unless converted into an approved clean SEO route with unique content.
?sort=price&view=grid&pageSize=100	No separate indexable page; canonical/noindex policy applied centrally.

URL / State	Behavior
Pagination	Crawlable paginated links. Infinite scroll may enhance but must not replace pagination.

9. Internal Linking Engine

The CMS/API must expose relationships that let templates create bounded contextual link modules. These relationships are the primary mechanism for crawl flow and entity understanding.

Source	Target	UI Module	Anchor Logic	Data Rule
Homepage	Commercial pages	Primary nav + commercial pathways	Use exact page H1 or concise commercial anchor	Always link P0 pages from raw HTML
Homepage	Origin hub + offer list	Featured origins/offer modules	Coffee origins / Current green coffee offer list	Always
Commercial page	Origin hub/detail	Origins we can supply	{Origin} coffee beans	Curated or actual sourcing capability
Commercial page	Offer list	Availability CTA	current green coffee offer list	Always when offer list active
Commercial page	Knowledge guide	Decision-support links	contextual anchor from section topic	Use page-specific guides only
Origin hub	Origin detail	Origin cards	{Origin} coffee beans	Every active origin
Origin detail	Lot detail	Available lots module	{Origin} {lot/grade/process}	Active lots only
Origin detail	Commercial page	RFQ/content links	wholesale coffee beans / specialty coffee beans	Use relevance by origin and audience
Offer list	Lot detail	Lot cards	{Lot Name} {Origin}	Active lots only
Lot detail	Origin detail	Origin context link	{Origin} coffee beans	Required
Lot detail	Commercial/RFQ	CTA modules	request sample / request quote	Required
Knowledge detail	One primary commercial target	CTA/inline link	Exact commercial target term	One primary target to avoid cannibalization
Knowledge detail	Related guides	Related reading	Guide title or intent-based anchor	3-5 max

10. Rendering, Metadata & Technical SEO

- Homepage, static commercial pages, guides and trust pages: SSG/ISR is suitable.
- Origin pages: ISR preferred so content and availability modules update without losing stable canonical routes.
- Offer list and lot details: SSR or inventory-aware ISR because active lots, availability and prices can change.
- Search/filter/cart/wishlist may use CSR under noindex policies, but canonical entity/detail links must remain crawlable.
- Every indexable route outputs one unique title, one meta description, one H1, one canonical, intended robots directive, main content and core links in raw/server HTML.
- Important links must use standard `<a href>`. Client-side router.push/onClick can enhance behavior, not replace crawlable links.
- Images must declare width/height or aspect ratio, use responsive sizes, and avoid >100KB unoptimized media where possible.

10.1 Metadata engine rules

Page Type	Title Rule	Meta Description Rule
Homepage	Hills Coffee Green Coffee Trading & Sourcing Partner	Introduce Hills Coffee as a B2B green coffee sourcing and trading partner for roasteries, cafés, factories and coffee businesses.
Commercial supplier page	{Primary Topic} for Coffee Businesses Hills Coffee	Source {primary topic lowercase} with Hills Coffee. Request samples, current availability, origin options and tailored B2B quotations.
Origin detail	{Origin} Coffee Beans Supplier for Roasteries Hills Coffee	Source {Origin} green coffee beans for commercial roasting and supply programs. Explore profiles, grades, availability and request a quote.
Offer list	Current Green Coffee Offer List Hills Coffee	Browse current green coffee lots by origin, grade, process and availability. Request samples or a tailored quotation from Hills Coffee.
Lot detail	{Lot Name} {Origin} Green Coffee Hills Coffee	Review {Lot Name} sourcing details, grade, process and availability. Request samples or a quotation from Hills Coffee.
Knowledge guide	{Guide Title} Hills Coffee Knowledge	Learn {guide topic} from a business sourcing perspective, then move to the relevant Hills Coffee commercial page or RFQ.

10.2 Structured data architecture

- Use JSON-LD as the default format and generate it from the same data source that renders visible HTML.
- Use a connected `@graph` on indexable pages so `WebPage/CollectionPage`, `BreadcrumbList`, `Organization`, `Service`, `Product/Offer`, `Place` and `Article` nodes can reference one another.
- Do not output hidden SEO-only schema values. Every material schema value must be visible, factual and current on the page.
- Use `Product + Offer` for individual coffee lots only when product/offer facts are visible and accurate. Commercial category pages should not be marked as a single `Product`.

- FAQPage is optional semantic markup only for page-specific visible Q&A.

Template / Page Type	Recommended Graph	Implementation Notes
Homepage	WebPage + WebSite + Organization	Single organization @id reused across site.
Commercial supplier page	WebPage + Service + BreadcrumbList + optional FAQPage	Service describes sourcing/wholesale function; facts must match visible page.
Origin hub/detail	CollectionPage/WebPage + Place + ItemList + BreadcrumbList	Origin/country entity plus visible lot/origin list where present.
Offer list hub	CollectionPage + ItemList + BreadcrumbList	ItemList mirrors rendered active lot cards only.
Coffee lot detail	Product + Offer + BreadcrumbList + Organization seller	Offer only when price/availability are visible and current.
Knowledge detail	Article/BlogPosting + Person + Organization + BreadcrumbList	Author, dates and sources mirror visible article metadata.
Contact/RFQ	ContactPage + Organization + ContactPoint/Service	Use verified contact details and sourcing CTA.

10.3 Stable @graph node model

Node	Stable ID Pattern	Relationship Pattern
Organization	https://www.hillscoffees.com/#organization	publisher/provider/seller for site, services, guides and coffee lot offers.
WebSite	https://www.hillscoffees.com/#website	publisher → #organization; url → canonical root.
WebPage / CollectionPage	{canonicalURL}#webpage	isPartOf → #website; breadcrumb → #breadcrumb; mainEntity → page entity.
BreadcrumbList	{canonicalURL}#breadcrumb	Matches visible breadcrumb hierarchy and canonical URLs.
Origin Place	/coffee-origins/{origin}/#origin	Referenced by origin page and related lot pages.
Coffee Lot Product	/green-coffee-offer-list/{slug-id}/#product	Referenced by WebPage.mainEntity and Offer.itemOffered.
Offer	/green-coffee-offer-list/{slug-id}/#offer	Contains visible current price/availability/URL/seller only when factual.
Article	/knowledge/{slug}/#article	mainEntityOfPage → canonical URL; author/reviewer/publisher relationships.

11. XML Sitemaps, Robots & Search / AI Discovery

```
/sitemap.xml
├── /sitemap-static.xml
├── /sitemap-commercial.xml
├── /sitemap-origins.xml
├── /sitemap-offer-list.xml
├── /sitemap-knowledge.xml
└── /sitemap-support.xml
```

- Only 200, indexable, canonical, published URLs enter XML sitemaps.
- Remove deleted, redirected, noindex, filtered, cart/wishlist and private URLs promptly.
- Use accurate lastmod when a page’s content, availability or product facts meaningfully change.
- robots.txt must reference the sitemap index and must not block critical rendering resources.
- Public SEO pages should be accessible to intended search crawlers and approved AI/search crawlers; CDN/WAF should not serve challenge pages to them.
- Use IndexNow for eligible add/update/delete events where the technical stack supports it, especially active offer-list changes.

12. Mobile, UX & Performance Requirements

Requirement	Development Expectation
First viewport	Identity, B2B proposition, key facts/availability and RFQ action are visible early. Do not bury commercial purpose under long SEO text.
Filters	Use accessible mobile drawer/sheet; no desktop-only filter bar assumption.
Interaction	No essential action depends on hover. Cards, links and form controls have touch-friendly targets.
Tables	Origin/lot/specification tables collapse into stacked or horizontally scrollable accessible layouts.
Images	Reserve dimensions/aspect ratios, use responsive sources, prioritize LCP image, lazy-load below fold.
Scripts	Defer maps/widgets/third parties until after primary text, cards and links are available.
Core Web Vitals	Test populated real templates, not empty staging pages.

13. Migration & Redirect Plan

Incoming Route / Pattern	Target / Behavior	HTTP	Reason / Notes
/products/	/green-coffee-offer-list/	301	Legacy product catalog consolidates to canonical B2B offer list.
/full-offer-list/	/green-coffee-offer-list/	301	Legacy offer list route consolidates to canonical offer list.
/products/{id}/	/green-coffee-offer-list/{origin-lot-slug}-{stable-id}/	301 when active mapping exists	Resolve by stable lot/product ID.
/coffee-origins/vietnam-robusta/	/coffee-origins/vietnam/	301 unless a unique Vietnam Robusta page is approved	Avoid duplicate origin intent.
/origins/	/coffee-origins/	301	Legacy origin hub route.
/en/origins/	/coffee-origins/	301	Legacy language route if English-only canonical site is used.
/spot-offerings/	/green-coffee-offer-list/	301	Legacy offerings intent.
/pays/{country-id}/	/coffee-origins/{mapped-origin}/	301 when exact country mapping exists	Old country URLs map to canonical origin pages.
/blog/{legacy-slug}/	/knowledge/{mapped-slug}/	301 when equivalent content exists	Old blog routes map one-to-one to new knowledge guides.
HTTP / non-preferred host	https://www.hillscoffees.com/{path}/	301	Canonical protocol/host in one hop.

14. Suggested Development Responsibility Structure

```

src/
├── app-or-routes/
│   ├── commercial/
│   ├── coffee-origins/
│   ├── green-coffee-offer-list/
│   ├── knowledge/
│   ├── request-a-quote/
│   └── trust-legal-utility/
├── templates/
│   ├── homepage/
│   ├── commercial-landing/
│   ├── origin-hub/
│   └── origin-detail/

```

- └─ offer-list/
- └─ coffee-lot-detail/
- └─ knowledge/
- └─ contact-trust/
- └─ components/
 - └─ navigation/ breadcrumbs/ rfq-forms/
 - └─ coffee-lot-cards/ offer-list/ filters/
 - └─ origin-cards/ facts-tables/ faq/
 - └─ author-reviewer/ trust/ related-links/
- └─ entities/
 - └─ origin/ coffee-lot/ commercial-page/ guide/ person/ organization/
- └─ seo/
 - └─ metadata/ canonical/ schema/ robots/ sitemap/ redirects/
- └─ analytics/
- └─ monitoring/
- └─ tests/
 - └─ seo-rendering/ schema/ redirects/ sitemap/ mobile/ performance/

15. Development Acceptance Criteria / Launch Gate

Area	Release Blocker Criterion
Rendering	Raw/server HTML contains title, meta description, H1, canonical, main content and core links for every indexable template.
HTTP	No soft 404s; lifecycle status codes match documented rules.
Crawlability	P0/P1 routes, origins, active lots and guides are reachable through standard anchors and pagination.
Indexation	Index/noindex/canonical rules match the URL matrix; filters and utility states do not pollute the index.
Sitemaps	Only 200, indexable, canonical, published URLs; accurate lastmod; no redirected/dead/filter URLs.
Schema	Valid JSON-LD describes visible factual content; breadcrumbs match canonical hierarchy.
Metadata	No [object Object] or duplicate/placeholder title/description values.
Mobile	Primary browse/filter/detail/RFQ flows work without hover or obstruction.
Performance	Critical content not blocked by large images, maps, sliders or third parties.
Crawler access	Public pages accessible through robots/WAF to intended search and AI crawlers.

Area	Release Blocker Criterion
Migration	Legacy URLs mapped; redirects one hop; internal links and sitemaps use new canonical URLs.
Governance	No unregistered indexable route outside the canonical URL matrix.

15.1 Implementation sequence

- Lock canonical route tree, route owners, entity IDs and redirect principles.
- Build entity/data model for origins, lots, commercial pages, guides, people and organization.
- Choose rendering strategy by template and fix API failure paths.
- Build shared navigation, breadcrumbs, cards, tables, RFQ, FAQ, author and related-link components.
- Build core templates: homepage, commercial pages, origins, offer list, lot detail, knowledge, RFQ/contact.
- Implement metadata, canonical, robots, sitemaps, structured data and lifecycle HTTP responses.
- Implement internal linking engine, pagination and filter indexation controls.
- Run staging crawl, schema validation, mobile/performance QA and redirect validation.
- Launch with monitoring and 7/14/30-day migration checkpoints.

16. Governance & Team Handoff

- This document is the authoritative architecture rulebook for the rebuild.
- Any new indexable route requires: page owner, search intent/keyword owner, parent, template, canonical/indexation rule, sitemap group and internal-link plan.
- SEO acceptance criteria belong in engineering QA and definition of done, not as a post-launch SEO checklist.
- Content team must write page-specific content for commercial and origin pages; templated synonyms alone are not sufficient.
- Dev team must not introduce new route aliases, filter pages or client-only links without SEO architecture approval.

Appendix A — Priority Route Summary

Priority	URL	Template	H1 / Page Name	Primary KW	Intent	Primary CTA
P0	/green-coffee-beans-supplier/	T02	Green Coffee Beans Supplier for Coffee Businesses	green coffee beans supplier	B2B supplier evaluation / sourcing inquiry	RFQ lead generation
P0	/coffee-beans-supplier/	T02	Coffee Beans Supplier for Roasteries, Cafés and Coffee Businesses	coffee beans supplier	Supplier discovery / commercial comparison	Speak with sourcing team
P0	/wholesale-coffee-beans /	T02	Wholesale Coffee Beans for Coffee Businesses	wholesale coffee beans	Wholesale buying / price inquiry	Wholesale quote

Priority	URL	Template	H1 / Page Name	Primary KW	Intent	Primary CTA
P0	/specialty-coffee-beans/	T02	Specialty Coffee Beans Sourcing for Coffee Businesses	specialty coffee beans	Specialty sourcing / high-value research	Specialty offer list
P0	/raw-coffee-beans-for-roasters/	T07	Raw Coffee Beans for Roasteries	raw coffee bean	Roastery procurement	Current offer list
P1	/bulk-coffee-beans/	T08	Bulk Coffee Beans for Commercial Buyers	bulk coffee beans	Bulk buying / logistics	Discuss bulk supply
P1	/coffee-beans-wholesale-price/	T08	Coffee Beans Wholesale Price Guide	coffee beans wholesale price	Commercial price investigation	Request latest price list
P1	/arabica-coffee-beans-wholesale/	T02	Arabica Coffee Beans Wholesale for Businesses	arabica coffee wholesale	Coffee type sourcing	Request Arabica quote
P1	/robusta-coffee-beans-wholesale/	T02	Robusta Coffee Beans Wholesale for Businesses	robusta coffee beans wholesale	Coffee type sourcing	Request Robusta quote
P0	/coffee-origins/	T05	Coffee Origins: Source Green Coffee Beans from the World's Best Coffee Growing Regions	green coffee beans	Informational + commercial investigation	Origin hub / RFQ path
P0	/coffee-origins/brazil/	T06	Brazil Coffee Beans Supplier for Roasteries & Coffee Businesses	brazil coffee beans	Commercial investigation	Brazil quote
P0	/coffee-origins/ethiopia/	T06	Ethiopian Coffee Beans Supplier for Specialty Roasteries	ethiopian coffee beans	Commercial investigation	Ethiopia samples
P0	/coffee-origins/colombia/	T06	Colombian Coffee Beans Supplier for Coffee Businesses	colombian coffee beans	Commercial investigation	Colombia quote
P1	/coffee-origins/vietnam/	T06	Vietnam Coffee Beans Supplier for Commercial Coffee Production	vietnam coffee beans	Commercial investigation	Vietnam Robusta quote
P1	/coffee-origins/uganda/	T06	Uganda Coffee Beans Supplier for Roasteries & Importers	uganda coffee beans	Commercial investigation	Uganda quote
P1	/coffee-origins/kenya/	T06	Kenya Coffee Beans Supplier for Specialty Coffee Roasteries	kenya coffee beans	Commercial investigation	Kenya quote

Priority	URL	Template	H1 / Page Name	Primary KW	Intent	Primary CTA
P0	/green-coffee-offer-list/	T03	Current Green Coffee Offer List	green coffee offer list	Active lot discovery	Request lot quote
P0	/green-coffee-offer-list/{origin-lot-slug}-{stable-id}/	T04	Individual Green Coffee Lot Detail	origin/lot-specific coffee	Exact offer investigation	Request sample / quote
P0	/request-a-quote/	T11	Request a Coffee Sourcing Quote	coffee sourcing quote	Lead conversion	Submit RFQ
P1	/about/	T12	About Hills Coffee	hills coffee	Brand/entity trust	Contact sourcing team
P1	/contact/	T11	Contact Hills Coffee	contact hills coffee	Contact / trust	Contact
P2	/help/	T13	Help Center	hills coffee help	Customer support	Support
P2	/shipping/	T13	Shipping Information	hills coffee shipping	Support	Support
P2	/returns/	T13	Returns & Refunds Policy	hills coffee returns	Support/legal	Support
P2	/privacy/	T13	Privacy Policy	privacy policy	Legal	None
P2	/cookies/	T13	Cookie Policy	cookie policy	Legal	None
P2	/terms/	T13	Terms of Service	terms of service	Legal	None
NOINDEX	/cart/	T14	Cart	cart	Private transaction state	Checkout
NOINDEX	/wishlist/	T14	Wishlist	wishlist	Private/user state	Save/compare
P1	/knowledge/	T09	Green Coffee Knowledge Center for Business Buyers	green coffee knowledge	Informational hub	Move to RFQ or commercial page
P1	/knowledge/what-are-green-coffee-beans/	T10	What Are Green Coffee Beans?	green coffee beans	Informational	Explore green coffee supply
P1	/knowledge/green-coffee-vs-roasted-coffee/	T10	Green Coffee vs Roasted Coffee for Business Buyers	green coffee vs roasted coffee	Informational / commercial	Choose supply model
P1	/knowledge/arabica-vs-robusta-commercial-buyers/	T10	Arabica vs Robusta for Commercial Coffee Buyers	arabica vs robusta	Informational / commercial	Ask for blend recommendation
P1	/knowledge/how-to-choose-green-coffee-supplier/	T10	How to Choose a Green Coffee Supplier	coffee beans wholesale suppliers	Commercial investigation	Supplier evaluation call

Priority	URL	Template	H1 / Page Name	Primary KW	Intent	Primary CTA
P1	/knowledge/how-roasteries-source-coffee-beans/	T10	How Roasteries Source Coffee Beans	coffee beans for roasters	Commercial investigation	Current offer list
P1	/knowledge/coffee-bean-grades-explained/	T10	Coffee Bean Grades Explained for Buyers	coffee bean grades	Informational	Specs-based quote
P1	/knowledge/coffee-defects-explained/	T10	Coffee Defects Explained: What Buyers Should Check	coffee defects	Informational	Sample review
P1	/knowledge/pallet-vs-container-coffee-buying/	T10	Pallet vs Container Coffee Buying	bulk coffee beans wholesale	Commercial investigation	Discuss shipping option

Appendix B — Developer Backlog Epics

Epic	Area	Objective
EPIC 01	Routing & URL Architecture	Implement canonical route tree, trailing slash policy, redirects and route ownership matrix.
EPIC 02	Entity/Data Model	Create Origin, Lot, CommercialPage, Guide, Person and Organization data structures.
EPIC 03	Rendering Framework	Set SSG/ISR/SSR/CSR responsibilities by template and ensure raw HTML output for indexable pages.
EPIC 04	SEO Metadata Engine	Generate title, meta description, canonical, robots, OG/Twitter and hreflang policy where applicable.
EPIC 05	Navigation & Breadcrumbs	Build crawlable header/footer navigation and BreadcrumbList-aligned breadcrumbs.
EPIC 06	Commercial Page Templates	Build T02/T07/T08 supplier, segment, pricing and bulk pages.
EPIC 07	Origin Templates	Build T05/T06 origin hub/detail templates and origin relationships.
EPIC 08	Offer List & Lot Detail	Build active offer list, coffee lot cards, lot detail lifecycle and RFQ actions.
EPIC 09	Knowledge Templates	Build knowledge hub/detail with author/reviewer, dates, sources and primary commercial target.

Epic	Area	Objective
EPIC 10	Internal Linking Engine	Generate bounded contextual links from entity relations.
EPIC 11	Faceted Navigation	Control filters, sort, pagination, canonical/noindex behavior.
EPIC 12	Structured Data	Implement JSON-LD @graph for Organization, Service, CollectionPage, Product/Offer, Article and breadcrumbs.
EPIC 13	Sitemaps & Robots	Create segmented XML sitemaps, robots.txt rules and crawler access checks.
EPIC 14	Migration & Monitoring	Implement one-hop redirects, launch crawl QA, alerting and Search Console monitoring.

Appendix C — QA Checklists

QA Area	Required Test
SEO QA	Crawl staging; verify canonical, metadata, H1s, noindex rules, pagination, internal links and sitemap eligibility.
Frontend QA	Validate responsive layouts, accessible navigation, crawlable anchors, no JS-only critical links, images with dimensions.
Backend QA	Validate entity IDs, lot lifecycle, 404/410/503 behavior, inventory API fallback and data freshness.
Structured Data QA	Validate JSON-LD, visible-data parity, stable @id reuse, breadcrumb match and no stale Offer data.
Redirect QA	Test legacy routes, old product IDs, host/protocol, trailing slash, one-hop redirects and no chains.
Sitemap QA	Inspect sitemap index and segmented files; verify only canonical 200 indexable URLs and accurate lastmod.
Crawler QA	Check robots.txt, WAF/CDN bot access and Search Console URL inspection for representative templates.
Content QA	Verify page-specific copy, FAQ relevance, author/reviewer/date visibility and no commercial-guide cannibalization.

QA Area	Required Test
Analytics QA	Track RFQ, sample request, offer list interaction, contact clicks and key navigation events without replacing links.